

1. Star Patterns

a. Square

```
import java.util.*;  
public class Square {  
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter number of rows  
to be printed: ");  
        int n = sc.nextInt();
```

```
        for (int i = 1; i <= n; i++) {  
            for (int j = 1; j <= n; j++) {  
                System.out.print("* ");
```

```
            }  
            System.out.println();
```

```
        }
```

```
    }
```

```
}
```

Enter the number of rows to be printed:

5

* * * * *

* * * * *

* * * * *

* * * * *

* * * * *

b. HollowSquare

```
import java.util.*;  
public class HollowSquare {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.println("Enter the number of  
        rows to be printed: ");  
        int n = sc.nextInt();  
        for (int i = 1; i <= n; i++) {  
            for (int k = 1; k <= n; k++) {  
                if (i == 1 || i == n || k == 1 || k == n) {  
                    System.out.print(" * ");  
                }  
                else {  
                    System.out.print("   ");  
                }  
            }  
            System.out.println();  
        }  
    }  
}
```

Enter the number of rows to be printed:

7

```
* * * * *
*           *
*           *
*           *
*           *
*           *
* * * * *
```

c. Triangle

```
import java.util.*;
```

```
public class Triangle {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter number of  
rows to be printed: ");
```

```
        int n = sc.nextInt();
```

```
        for (int i = 1; i <= n; i++) {
```

```
            for (int j = 1; j <= i; j++) {
```

```
                System.out.print("* ");
```

```
            }
```

```
            System.out.println();
```

```
        }
```

```
    }
```

Enter the number of rows to be printed:

5

*

* *

* * *

* * * *

* * * * *

d. Inverted Triangle / Half Diamond Inverted

```
import java.util.*;  
public class InvertedTriangle {  
    public static void main (String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter the number of  
        rows to be printed: ");  
        int n = sc.nextInt();
```

```
        for (int i = 1; i <= n; i++) {  
            for (int space = 0; space < i; space++) {  
                System.out.print(" ");  
            }
```

```
            for (int j = 0; j < (2 * n - (2 * i + 1)); j++) {  
                System.out.print("* ");  
            }  
            System.out.println();
```

```
        }  
    }  
}
```

Enter the number of rows to be printed:

5

* * * * *

* * * *

* *

*

e. Pyramid.

```
import java.util.*;
public class Pyramid {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the number of rows  
to be printed : ");
        int n = sc.nextInt();

        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= n - i; j++) {
                System.out.print(" ");
            }
            for (int k = 1; k <= (2 * i - 1); k++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

Enter the number of rows to be printed:

5

*

1. Diamond

```
import java.util.*;
```

```
public class Diamond{
```

```
    public static void main(String[] args){
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter the number of rows  
        to be printed: ");
```

```
        int n = sc.nextInt();
```

```
        for (int i=1; i<=n; i++){  
            for (int j=1; j<=n-i; j++){  
                System.out.print(" ");  
            }
```

```
            for (int k=1; k<=(2*i-1); k++){  
                System.out.print("*");  
            }  
            System.out.println();  
        }
```

```
        for (int i=n-1; i>=1; i--){  
            for (int j=1; j<=n-i; j++){  
                System.out.print(" ");  
            }
```

```
            for (int k=1; k<=(2*i-1); k++){  
                System.out.print("*");  
            }  
            System.out.println();  
        }
```

```
    }
```

Enter the number of rows to be printed:

5

*

*

g. HalfDiamond

```
import java.util.*;
```

```
public class HalfDiamond{
```

```
    public static void main(String[] args){
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter the number of rows  
to be printed: ");
```

```
        int n = sc.nextInt();
```

```
        for(int i=1; i<=n; i++){  
            for(int j=1; j<=i; j++){  
                System.out.print("*");
```

```
            }
```

```
            System.out.println();
```

```
        }
```

```
        for(int i=n-1; i>=1; i--){  
            for(int j=1; j<=i; j++){  
                System.out.print("*");
```

```
            }
```

```
            System.out.println();
```

```
        }
```

```
    }
```

```
}
```

Enter the number of rows to be printed:

6

*

**

**

*

2. Number Patterns

a. Square

```
import java.util.*;
public class Square {
    public static void main (String [] args) {
        Scanner sc = new Scanner (System.in);

        System.out.println("Enter the number of
        rows to be printed: ");
        int n = sc.nextInt();

        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= n; j++) {
                System.out.print ("1");
            }
            System.out.println();
        }
    }
}
```

Enter the number of rows to be printed:

6

1 1 1 1 1 1

1 1 1 1 1 1

1 1 1 1 1 1

1 1 1 1 1 1

1 1 1 1 1 1

1 1 1 1 1 1

6. Incrementing Square

```
import java.util.*;
```

```
public class Square {  
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter the number of  
rows to be printed:");  
        int n = sc.nextInt();
```

```
        for (int i = 1; i <= n; i++) {  
            for (int j = 1; j <= n; j++) {  
                System.out.print(j + " ");  
            }  
            System.out.println();  
        }
```

```
    }  
}
```

Enter the number of rows to be printed:

6

1 2 3 4 5 6

1 2 3 4 5 6

1 2 3 4 5 6

1 2 3 4 5 6

1 2 3 4 5 6

1 2 3 4 5 6

C. Triangle.

```
import java.util.*;  
public class Triangle {  
    public static void main (String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.println("Enter the number of  
rows to be printed: ");  
        int n = sc.nextInt();  
        for (int i = 1; i <= n; i++) {  
            for (int j = 1; j <= i; j++) {  
                System.out.print(j + " ");  
            }  
            System.out.println();  
        }  
    }  
}
```

Enter the number of rows to be printed:

5

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

d. Inverted Diamond

```
import java.util.*;
```

```
public class InvertedDiamond {  
    public static void main (String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter the number of rows  
        to be printed : ");  
        int n = sc.nextInt();
```

```
        for (int i = n; i >= 1; i--) {  
            for (int j = 1; j <= n - i; j++) {  
                System.out.print(" ");  
            }
```

```
            for (int k = 1; k <= i; k++) {  
                System.out.print(i + " ");  
            }
```

```
            System.out.println();  
        }
```

```
    }  
}
```

Enter the number of rows to be printed:

5

5 5 5 5 5

4 4 4 4

3 3 3

2 2

1

e. Sandwich

```
import java.util.*;
public class Sandwich {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the number of
        rows to be printed: ");
        int n = sc.nextInt();
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j < i; j++) {
                System.out.print(" ");
            }
            for (int k = i; k <= n; k++) {
                System.out.print(i + " ");
            }
            System.out.println();
        }
        for (int i = n - 1; i >= 1; i--) {
            for (int j = 1; j < i; j++) {
                System.out.print(" ");
            }
            for (int k = i; k <= n; k++) {
                System.out.print(i + " ");
            }
            System.out.println();
        }
    }
}
```

Enter the number of rows to be printed:

5

1 1 1 1 1

2 2 2 2

3 3 3

4 4

5

4 4

3 3 3

2 2 2 2

1 1 1 1 1

f. HalfDiamond

```
import java.util.*;
```

```
public class HalfDiamond{
```

```
    public static void main (String [ ] args){
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter the numbers of  
        rows to be printed: ");
```

```
        int n = sc.nextInt();
```

```
        for (int i=1; i <= n; i++){
```

```
            for (int j=1; j <= i; j++){
```

```
                System.out.print (i + " ");
```

```
            }
```

```
            System.out.println ( );
```

```
        }
```

```
        for (int i = n-1; i >= 1; i--){
```

```
            for (int j=1; j <= i; j++){
```

```
                System.out.print (i + " ");
```

```
            }
```

```
            System.out.println ( );
```

```
        }
```

```
    }
```

```
}
```

Enter the number of rows to be printed:

4

1

2 2

3 3 3

4 4 4 4

3 3 3

2 2

1